

Odyssey

AI Travel Planner

Framework: OpenAI Agents SDK + Streamlit | Version: 1.0 | Date: June 2026
Repository: bibhu2020/agenticaai (Hugging Face Space)

1. Overview

Odyssey is a structured AI travel planner. Users describe a trip (destination, budget, preferences) and the system returns a complete TravelPlan including flight options, hotel options, weather remarks, activities, and travel notes. A budget guardrail agent intercepts unrealistic budget inputs before the main agent runs. Built on the OpenAI Agents SDK with structured output types.

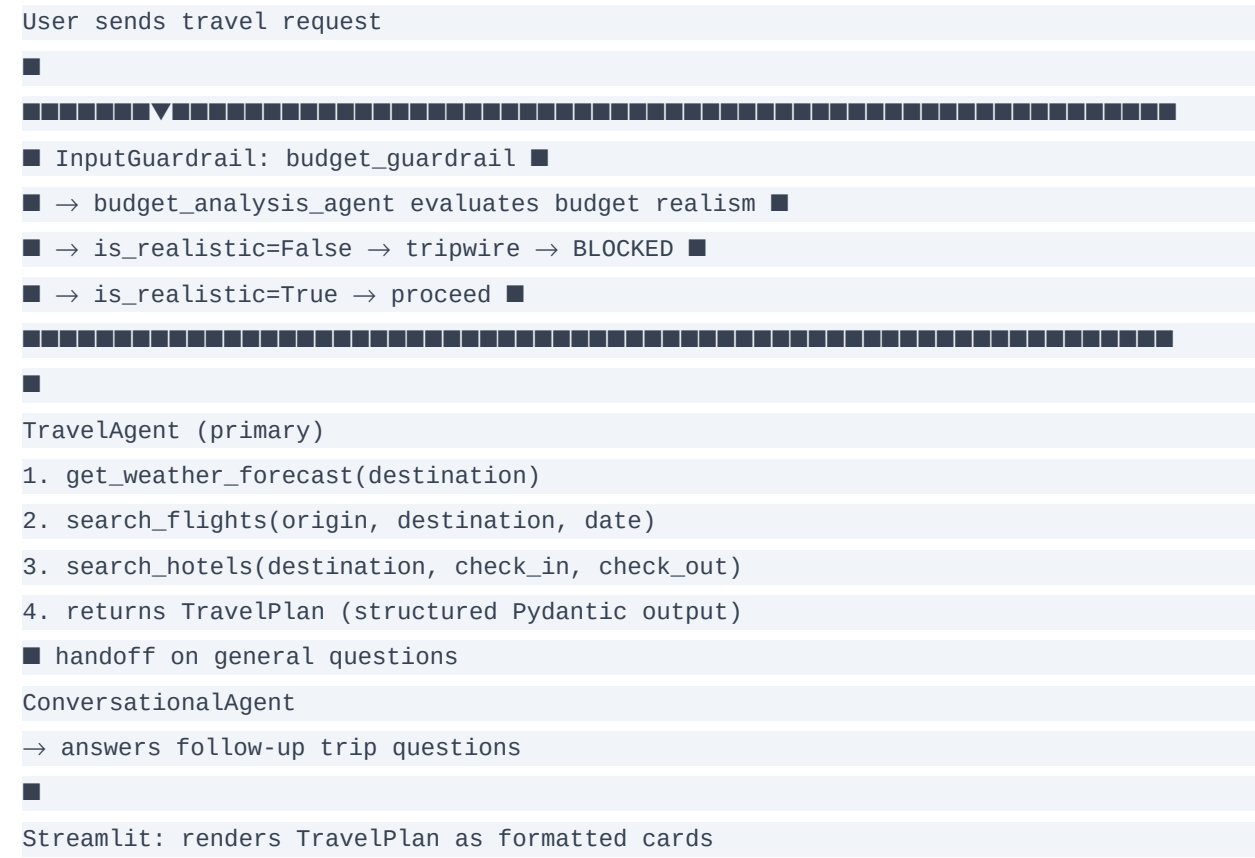
2. System Design

2.1 Components

Component	Technology	Purpose
app.py	Streamlit	Chat UI with TravelPlan structured display
aagents/travel_agent.py	OpenAI Agents SDK Agent	Primary agent: calls tools, returns TravelPlan
aagents/conversational_agent.py	OpenAI Agents SDK Agent	Handles general trip questions via handoff
aagents/budget_guardrail_agent.py	OpenAI Agents SDK Agent	Budget validation via InputGuardrail tripwire
tools/weather_mock.py	function_tool	Returns weather forecast for destination
tools/flight.py	function_tool	Searches available flight options
tools/hotel.py	function_tool	Searches available hotel options

Component	Technology	Purpose
tools/search.py	function_tool	Web search for activities, attractions
output_types/travel_plan.py	Pydantic BaseModel	Structured output schema: flights, hotels, weather, activities
output_types/budget_analysis.py	Pydantic BaseModel	Guardrail output: is_realistic, reasoning
contexts/user_context.py	dataclass	Per-run user context injected into all agents
llm.py	openai.AsyncOpenAI	LLM factory (openai / google)

2.2 Architecture Diagram



3. Agentic Flow

Step	Actor	Action
1	Budget Guardrail	Assess if user's stated budget is realistic; tripwire if wildly unrealistic
2	TravelAgent	Call get_weather_forecast for destination
3	TravelAgent	Call search_flights for route and date
4	TravelAgent	Call search_hotels for stay details
5	TravelAgent	Compile all results → return TravelPlan structured object

Step	Actor	Action
6	ConversationalAgent	Respond to any follow-up general questions (via handoff)
7	Streamlit	Display TravelPlan cards and/or conversational response

4. Structured Output Types

Type	Fields	Producer
TravelPlan	flight_options, hotel_options, weather_remark, activities, notes	TravelAgent
BudgetAnalysis	is_realistic (bool), reasoning (str)	budget_analysis_agent
FlightRecommendation	airline, price, duration, departure_time	TravelAgent via search_flights
HotelRecommendation	name, price_per_night, rating, amenities	TravelAgent via search_hotels

5. Configuration

Env Var	Required for
OPENAI_API_KEY	Default LLM (gpt-4o-mini)
GOOGLE_API_KEY	Gemini model option